OPERATION HATA

Hidden Attachment Threat Analyzer — Complete Setup & Installation Guide

Operation HATA is an AI-assisted cybersecurity platform that automatically monitors email inboxes for image attachments, isolates them in a secure quarantine environment, runs them through a multi-layer 18-stage forensic analysis pipeline (entropy, steganography, YARA rules, OCR, QR heuristics, and AI report generation), and visualizes threat metrics on a real-time SOC dashboard.

1. System Requirements & Prerequisites

Before setting up Operation HATA, ensure your system has the following software installed:

- **Operating System:** Windows 10/11, macOS, or Ubuntu Linux (20.04+).
- **Python:** Version 3.11 or higher (with pip enabled).
- **Node.js:** Version 18.0 or higher (with npm 9+).
- **Git:** Installed and available in terminal.
- **(Optional Forensic Tools):** Tesseract OCR (for image text detection) and ExifTool.

2. Clone the Repository

Open PowerShell, Command Prompt, or Terminal, and run:

```
git clone https://github.com/Crypton-ctrl/Operation-HATA.git
cd Operation-HATA/hata
```

3. Backend Setup (FastAPI & Python)

The backend server powers the API, database, analysis pipeline, and 24/7 email monitoring loop.

Step 3.1: Create & Activate Virtual Environment

Navigate to the backend directory and set up a clean Python virtual environment:

```
cd backend
# Create virtual environment
python -m venv venv

# Activate on Windows (PowerShell):
.\venv\Scripts\Activate.ps1

# OR activate on Linux / macOS:
source venv/bin/activate
```

If Windows displays a script execution policy error, run: Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass before activating.

Step 3.2: Install Python Dependencies

```
pip install -r requirements.txt
```

Step 3.3: Configure the .env File

Copy the example configuration to create your active .env file:

```
# Windows (PowerShell):
Copy-Item .env.example .env

# Linux / macOS:
cp .env.example .env
```

Open .env in your text editor. Key variables to verify or set:

Variable	Description & Default Value
SECRET_KEY	Random secret key for token encryption (e.g. hata-secret-key-2026)
CORS_ORIGINS	http://localhost:5173,http://localhost:5174,http://localhost:5175,https://operation-hata.netlify.app
DATABASE_URL	sqlite:///./hata.db (zero-config, local database)
GMAIL_CLIENT_ID	Your Google OAuth Web Client ID (from Google Cloud Console)
GMAIL_CLIENT_SECRET	Your Google OAuth Client Secret
GMAIL_REDIRECT_URI	http://localhost:8000/api/emails/oauth/callback

Step 3.4: Launch the Backend Server

```
# From inside the 'backend' folder:
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload
```

You will see: Uvicorn running on http://127.0.0.1:8000. The SQLite database (hata.db) will be created automatically on startup!

4. Frontend Setup (React + Vite)

Open a **new terminal window** (keep the backend terminal running).

Step 4.1: Install Node Dependencies

```
cd Operation-HATA/hata/frontend
npm install
```

Step 4.2: Start the Frontend Dev Server

```
npm run dev
```

Vite will start the dev server (typically on **http://localhost:5174** or **http://localhost:5175**). Open this URL in your web browser to view the application!

5. Google OAuth 2.0 & Gmail Monitoring Setup

Operation HATA allows 1-click Google Sign-in that automatically connects the user's Gmail account to 24/7 mailbox threat analysis.

Step 5.1: Enable the Gmail API

1. Open the **Google Cloud Console** (https://console.cloud.google.com).
2. Navigate to **APIs & Services** → **Library**.
3. Search for **Gmail API** and click **ENABLE**.

Step 5.2: Configure OAuth Consent Screen & Test Users

1. Go to **APIs & Services** → **OAuth consent screen** (or **Audience**).
2. Select **External** User Type.
3. Set App Name to Operation HATA and enter your support email.
4. Leave *Application home page*, *Privacy policy*, and *Authorized domains* **BLANK**.
5. Under **Test users**, click **+ ADD USERS** and type your friend's Gmail address (and your own). Click **SAVE**.

Step 5.3: Create Web Client Credentials

1. Go to **APIs & Services** → **Credentials**.
2. Click **+ CREATE CREDENTIALS** → **OAuth client ID**.
3. Application type: **Web application**.
4. **Authorized JavaScript origins:** Add:
   - http://localhost:5173
   - http://localhost:5174
   - http://localhost:5175
   - https://operation-hata.netlify.app
5. **Authorized redirect URIs:** Add:
   - http://localhost:8000/api/emails/oauth/callback
   - http://127.0.0.1:8000/api/emails/oauth/callback
6. Click **CREATE** and copy the generated **Client ID** and **Client Secret** into your backend/.env file.

> ***TEST USERS VS PRODUCTION:*** *While your Google Cloud app is in Testing mode, only Gmail accounts added to 'Test users' can log in. To allow ANY person worldwide to log in, simply click 'PUBLISH APP' on the OAuth consent screen!*

6. Using Operation HATA

A. Logging In:

Open http://localhost:5175/login and click **Sign in with Google**. Upon signing in, your dynamic name and registered Gmail address will appear on the dashboard.

(Alternative Local Admin Account): You can also log in with Username: admin and Password: Admin@12345.

B. Automatic Gmail Monitoring:

Click **Automatic Monitoring** in the left sidebar. The system checks for unread emails with attachments every 60 seconds. When an image attachment arrives, it is isolated in quarantine, analyzed through the forensic pipeline, and an alert is broadcast to the dashboard in real-time.

C. Manual Threat Scanning:

Click **Manual Scan**. Drag and drop any image file (PNG, JPG, BMP, TIFF, GIF, WEBP) to run the 18 forensic checks: Magic bytes verification, SHA-256 hash, EXIF metadata, Shannon entropy, Steganography chi-square test, appended ZIP/PE payload detection, and QR code inspection.

D. Quarantine & PDF Reports:

View the **Quarantine** tab to inspect suspicious files safely. Click any completed scan to download a full, executive-ready forensic PDF report.

7. Troubleshooting & FAQ

Issue / Error	Root Cause & Fix
`Error 403: access_denied`	The Gmail ID is not listed under **Test users** in Google Cloud Console. Go to *Audience* → *Test users* → *+ ADD USERS*.
`Error 400: redirect_uri_mismatch`	The Authorized Redirect URI is missing. In Google Cloud Console Credentials, add http://localhost:8000/api/emails/oauth/callback with no trailing slash.
`Gmail API disabled (403)`	Enable the Gmail API in your Google Cloud Project: *APIs & Services* → *Library* → *Search 'Gmail API'* → *Click Enable*.
`Port 5174 is in use`	Vite automatically increments to port 5175. Check terminal output for the active URL and open http://localhost:5175.
`Could not connect to server on Netlify`	Netlify only hosts the frontend. For Netlify to work, deploy the backend to Render or AWS EC2 and add VITE_API_BASE_URL in Netlify Environment Variables.
`ExecutionPolicy error in PowerShell`	Run Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass in PowerShell before running Activate.ps1.

8. Quick Command Cheat Sheet

Keep this handy whenever starting the project:

```
# Terminal 1: Backend
cd Operation-HATA/hata/backend
.\venv\Scripts\activate
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload

# Terminal 2: Frontend
cd Operation-HATA/hata/frontend
npm run dev

# Access in Browser:
Frontend:  http://localhost:5175 (or 5174)
Backend Swagger Docs: http://localhost:8000/docs
```

Need help or want to contribute? Visit the project repository at: https://github.com/Crypton-ctrl/Operation-HATA